



Southeast University

Department of Computer Science and Engineering

LAB Report

Course Name: Computer Networking LAB

Course Code: CSE 342.2

Experiment No.: 08

Experiment Name: Dynamic Routing – Single Area OSPF Configuration and Protocol Analysis.

Date of performance: 16.09.26

Date of Submission: 23.09.26

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Experiment Name: Dynamic Routing - Single Area OSPF Configuration and Analysis.

Introduction: This lab experiment transitions the network infrastructure from distance-vector routing to link-state routing Paradigm using OSPF 1. OSPF utilizes Dijkstra's Shortest Path First (SPF) algorithm to calculate the most efficient route to network ~~distance~~ destinations by assessing link bandwidth (cost). By deploying all routers into a single backbone area (Area 0), the lab demonstrates how OSPF establishes neighbor adjacencies, synchronizes link-state databases and achieves superior Path Selection over mixed-media links (gig and serial) compared to legacy Protocols.

Objectives:

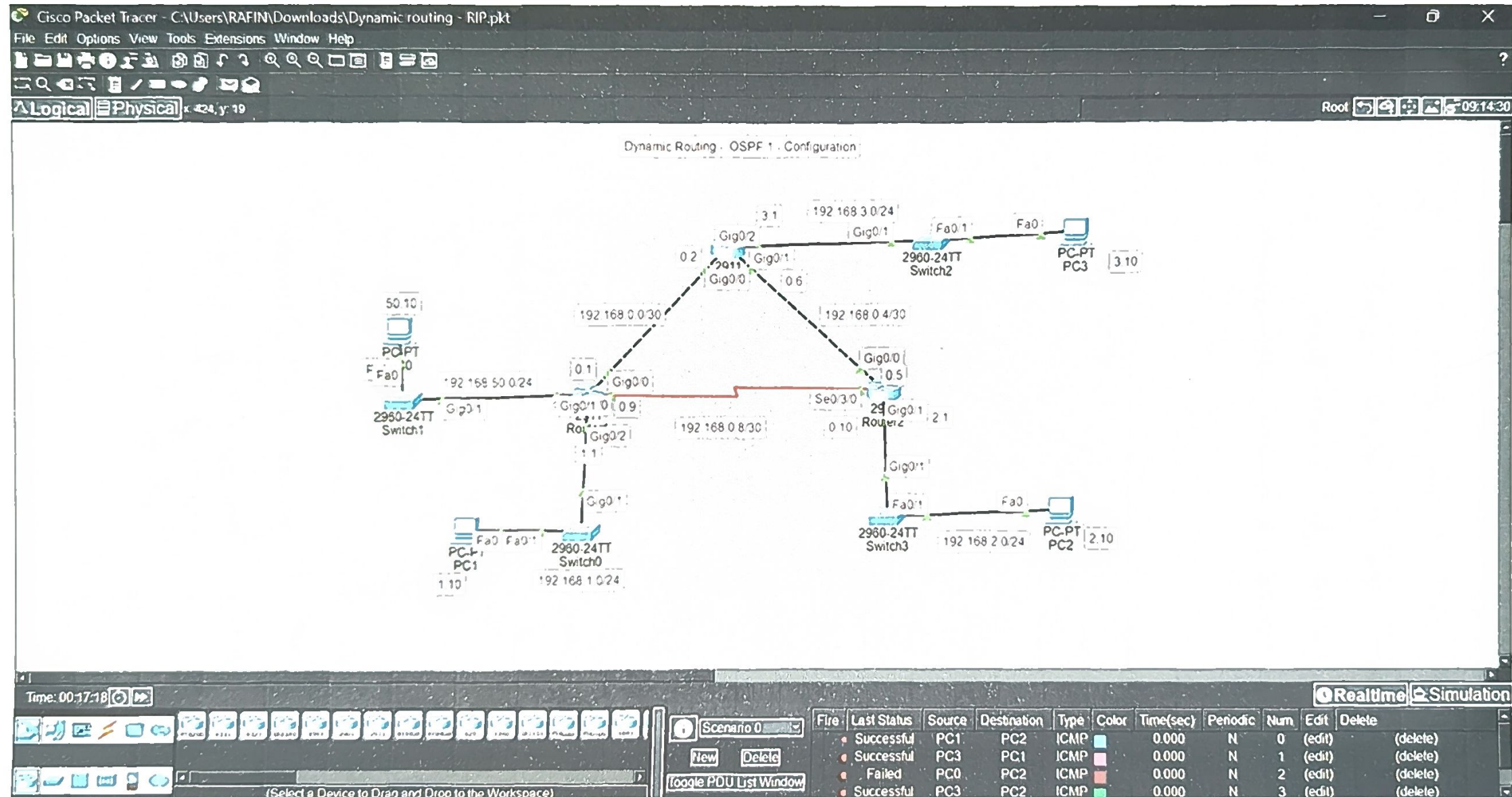
- To implement and verify Single-Area Open Shortest Path First (OSPF) dynamic routing across a multi-router enterprise topology
- To utilize wildcard masks for accurate network advertisement within OSPF Area 0.
- To critically analyze OSPF behavior, specifically focusing on Link-State Advertisement, Cost-based metric calculation and convergence efficiency compared

to distance-vector Protocols.

- To Perform end-to-end PDU simulation tests and troubleshoot any reachability anomalies.

Network Addressing Table:

Device	Interface	IP Address / subnet mask	Default Gateway
R1	Gig 0/1	192.168.1.1 / 24	N/A
R1	Gig 0/0	192.168.50.1 / 24	N/A
R1	Gig 0/2	192.168.0.1 / 30	N/A
R1	Se 0/3/0 (DCE)	192.168.0.9 / 30	N/A
R0	Gig 0/0	192.168.0.2 / 30	N/A
R0	Gig 0/2	192.168.3.1 / 24	N/A
R0	Gig 0/1	192.168.0.6 / 30	N/A
R2	Gig 0/1	192.168.2.1 / 24	N/A
R2	Gig 0/0	192.168.0.5 / 30	N/A
R2	Se 0/3/0 (DTE)	192.168.0.10 / 30	N/A
PC1	Fa 0	192.168.1.10 / 24	192.168.1.1
PC0	Fa 0	192.168.50.10 / 24	192.168.50.1
PC2	Fa 0	192.168.2.10 / 24	192.168.2.1
PC3	Fa 0	192.168.3.10 / 24	192.168.3.1



▣ Device Configuration Specifications: The following configuration establish single-Area OSPF using Process ID 1. Hence the required Subnet mask is Inverted to wildcard masks for network Advertisement.

• Router 1:

Router > Enable

Router# configure terminal

Router(config)# router ospf 1

Router(config)# Network 192.168.1.0 0.0.0.255 area 0

Router(config)# Network 192.168.50.0 0.0.0.255 area 0

Router(config)# Network 192.168.0.0 0.0.0.3 area 0

Router(config)# Network 192.168.0.8 0.0.0.3 area 0

Router(config)# End

• Router 2:

Router > Enable

Router# configure terminal

Router(config)# router ospf 1

Router(config)# Network 192.168.3.0 0.0.0.255 area 0

Router(config)# Network 192.168.0.0 0.0.0.3 area 0

Router(config)# Network 192.168.0.4 0.0.0.3 area 0

Router(config)# End

- Router 2:

Router > Enable

Router# Configure Terminal

Router(config)# Router OSPF 1

Router(config)# Network 192.168.2.0 0.0.0.255 area 0

Router(config)# Network 192.168.0.4 0.0.0.3 area 0

Router(config)# Network 192.168.0.8 0.0.0.3 area 0

Router(config)# End

Protocol Analysis for OSPF:

- Information sharing mechanisms: Unlike distance-vector Protocols, OSPF does not Periodically broadcast its entire routing table. Instead, it forms adjacencies with neighboring routers and exchanges Link-state Advertisements (LSAs) via multicast 224.0.0.5. once the link-state database is fully synchronized ~~across~~ across area 0, OSPF remains silent only sending triggered, Partial updates when a topology change occurs. This significantly conserves network bandwidth.
- Path Calculation and metric selection: OSPF Calculates optimal network Paths utilizing 'Cost' as its metric/ derived from the interface bandwidth. In ~~the~~ this topology, OSPF accurately calculates that the Gigabit Ethernet Path via the top router has a

significantly lower cumulative cost than the 1-hop direct Point-to-Point serial link. Consequently, OSPF smartly routes traffic over the high-speed gigabit links.

🔧 • Network Convergence Time: Convergence in OSPF is ~~ext~~ exceptionally rapid. Upon detecting a link failure (a physical down state), the connected router immediately generates a triggered LSA. All routers in Area 0 receive this update, instantly return the SPF Algorithm and update their routing tables to reflect the new optimal paths. This bypasses the slow invalidation timers used in legacy protocols, preventing prolonged packet loss during failure scenarios.

🔧 Verification: To verify the connectivity and OSPF database.

show ip ospf neighbors

show ip ospf database

show ip route